

SKILLS

- **Languages:** Python, C/C++, SQL
- **Machine Learning (ML) & Deep Learning (DL):** PyTorch, Scikit-learn, NumPy, Pandas
- **Generative AI & Retrieval (RAG):** Gemini API, Voyage AI (embeddings, rerank), Vector Search, HNSWlib, FAISS
- **Computer Vision (CV):** OpenCV, FaceNet / Inception-ResNet, YuNet, SFace
- **Reinforcement Learning (RL):** Gymnasium, PPO
- **Tools & Platforms:** Git, Docker, Linux, AWS, SQLite

EDUCATION

- **VIT Bhopal** Bhopal, MP
Bachelor of Technology - Computer Science and Engineering (AIML); CGPA: 8.4/10.0 Aug 2023 – May 2027

PROJECTS

- **Multi-Lane Hybrid Retrieval & Recommendation Engine** Jan 2026 – Present
Information Retrieval & Machine Learning Systems Python, SQLite, Gemini API, Voyage AI, HNSWlib
 - **Semantic Coverage:** Expanded semantic coverage from 0% to 100% across 10,659 long-tail films, scaling a 1,024-dim Voyage-4 HNSW index from 9,526 to 20,185 entries (+111.9%) via Gemini attribute extraction and k -NN label propagation over 52,897 Wikipedia plots and 25M+ ratings.
 - **Hybrid Retrieval Architecture:** Architected a three-lane retrieval system (dense HNSW, tag-genome HNSW, BM25) fused via Reciprocal Rank Fusion ($k = 60$), rescuing 100% of single-lane zero-recall blind spots and widening candidate diversity by +141.4%.
 - **Reranking & Efficiency:** Benchmarked a 369.7ms mean reranking latency and an 83.2% cut in LLM ranking tokens by integrating Voyage Rerank-2.5 over 100-candidate pools and replacing in-context LLM ranking with a structured query-understanding layer.
- **Prince of Persia Autonomous RL Agent & Environment (SDLPoP)** Oct 2025 – Present
Reinforcement Learning & Systems Engineering Python, PyTorch, Gymnasium, PPO, ctypes
 - **RL Environment & Throughput:** Built the first Gymnasium-compatible RL environment for Prince of Persia, interfacing Python with the decompiled SDLPoP C engine via ctypes and sustaining 13,167 steps/sec across 27 parallel workers.
 - **Memory Architecture & Task Mastery:** Achieved 100% Level 1 completion (95% stochastic) by pairing a 12-channel tile observation space with a 24-room topological state encoder, trained via PPO and FiGAR over 149.9M frames.
 - **Engine Verification & Reward Design:** Engineered a hierarchical reward system combining extrinsic milestone shaping (sword pickup, guard combat, level exit), intrinsic state curiosity, and memorized return-path routing to solve multi-room exploration and backtracking across Level 1.
- **Real-Time Facial Recognition Attendance System** Jan 2025 – Mar 2025
Computer Vision & Machine Learning Python, OpenCV, ONNX Runtime, FAISS, scikit-learn
 - **Verification Pipeline:** Built a real-time facial recognition attendance app in Python (CustomTkinter, OpenCV), achieving 14.7ms end-to-end latency across a 320-identity gallery.
 - **Recognition Accuracy:** Combined a YuNet detector with an SFace extractor (128-dim embeddings), validating 98.0% accuracy on LFW and 98.3% on Olivetti Faces via 5-fold cross-validation.
 - **Vector Search & Reliability:** Cut biometric matching latency to 8.6 μ s per query (3.7 $\times$ over brute-force NumPy) via FAISS L2 indexing.

LEADERSHIP EXPERIENCE

- **Tech Co-Lead – Bit by Bit VIT Bhopal** 2025 – 2026
Campus Technical Event Organization
 - **Technical Leadership:** Led a 10-member technical team as Tech Co-Lead, driving website development and technical infrastructure for club events, including a 200+ participant hackathon and senior/industry expert talks, from challenge delivery platforms to event registration systems.

CERTIFICATIONS AND BADGES

- **AWS Certified Cloud Practitioner (CLF-C02):** Amazon Web Services, 2025
- **Generative AI using IBM Watsonx:** IBM Career Education Program, Jun 2025

ACCOMPLISHMENTS

- Top 10 Finalist – National Cyber Shield Hackathon, MP Police & ClearTrail (SBI-sponsored), 2025
- Top 10 Finish – Hackcrux Hackathon, LNMIIT Jaipur, Mar 2025
- Merged Open-Source Contribution – JAX (Google’s ML Framework), Bug Fix #31101 (PR #31192), Sep 2025